

PageRank-Based Vaccination Strategies

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1 Introduction

The spread of epidemics, such as diseases or computer viruses, presents critical challenges in networks [16, 19]. Effective immunization strategies aim to vaccinate a select subset of nodes to minimize this spread [20, 5], especially when resources are scarce, as seen during the early stages of pandemics like COVID-19.

In this work, we tackle the problem of *adaptive vaccination*, focusing on selecting the optimal nodes for immunization based on the current state of infection spread, for times of scarce amounts of vaccines. We hypothesized that **PageRank**'s centrality, a well-established measure for identifying influential nodes, could effectively guide vaccination efforts. PageRank has traditionally been applied to web ranking [17], but we extend its use to vaccination strategy optimization.

We evaluate the performance of Random, Degree-based, PageRank-based, and NetShield algorithms [24] under both the Independent Cascade (IC) and Susceptible-Infected-Recovered (SIR) models [9, 10]. Our results on real-world networks demonstrate that the PageRank-based strategy consistently outperforms other heuristics in minimizing epidemic spread.

2 Problem Formulation

The goal of this work is to design a vaccination strategy that adapts to both the network's structure and the current state of an ongoing epidemic. Given a network $G=(V,E)$, where V represents the set of nodes and E represents the set of edges, let $S \subseteq V$ denote the set of infected nodes at a particular point in time. With a limited vaccination budget of k , our objective is to identify a set of uninfected ('healthy') nodes, denoted by $\mathcal{H}$, to vaccinate in order to minimize future infection spread.

We define the adaptive vaccination problem as finding a set $\mathcal{H} \subseteq V \setminus S$ of size $|\mathcal{H}|=k$ such that the expected number of infected nodes at the end of the epidemic, denoted as $\mathbb{E}_G^{S,\mathcal{H}}$, is minimized. This can be formalized as:

$$\mathcal{H}^* = \arg \min_{\mathcal{H} \subseteq V \setminus S, |\mathcal{H}|=k} \mathbb{E}_G^{S,\mathcal{H}}.$$

The challenge of this problem stems from its combinatorial nature, making it computationally intractable for large networks. Hence, we explore heuristic strategies that leverage network centrality measures to guide the vaccination process.

3 Background

Given that our problem formulation from Section 2 allows us to now reason about the spread of diseases in human populations and computer viruses in digital networks, we provide a brief overview of the relevant concepts and models.

3.1 Epidemic Models

We consider two epidemic models, each suited to different disease dynamics:

Susceptible-Infected-Recovered (SIR) Model [10] The SIR model is commonly used to study disease spread in networks [2, 7, 23]. It models infectious diseases with varying recovery rates, and we thus use it to model the low-recovery rate disease like Ebola, known for its high mortality. Specifically, infected nodes can transmit the disease to susceptible neighbors with a high probability $p=0.8$, and recover with a low probability $\delta=0.1$, becoming immune and ceasing transmission.

Independent Cascade (IC) Model [9] A variant of the SIR model, the IC model is used to simulate information diffusion in networks [9]—by allowing nodes to infect neighbors only once. We use it to model diseases like COVID-19, where quick quarantine and recovery are common. Specifically, infected nodes infect susceptible neighbors with probability $p=0.4$, but can only attempt infection once before becoming inactive.

3.2 Vaccination Strategies

Vaccination strategies in networks have been extensively studied [5, 8]. Preemptive immunization methods often rely on network centrality measures, such as degree or betweenness, to select nodes for vaccination [11]. Instead, adaptive approaches consider the current state of the epidemic to make vaccination decisions [18].

The NetShield algorithm [24] aims to select nodes that minimize the spectral radius of the adjacency matrix, thereby reducing the network's vulnerability to epidemic spread. Our work differs in that we explore the use of PageRank centrality as a heuristic for data-aware vaccination without prior knowledge of such algorithms.

4 PageRank Centrality

PageRank is a measure of node importance based on the idea that connections from important nodes contribute more to the importance of a node [17]. The PageRank score $PR(i)$ of node i is defined recursively as:

$$PR(v) = \frac{\epsilon}{N} + (1-\epsilon) \sum_{(v',v) \in E} \frac{PR(v')}{\text{out-degree}(v')}$$

where ϵ is the damping factor (typically set to $1/7$), and N is the total number of nodes. We aim to leverage PageRank to identify nodes that are both well-connected and central in the network structure for vaccination, as we believe these nodes are more likely to influence the spread of epidemics.

5 Methodology

5.1 Vaccination Strategies

We implement and compare the following strategies:

Strategy	Description	Function in Code
No Vaccination	No nodes are vaccinated.	N/A
Random	Randomly select k healthy nodes to vaccinate.	<code>random_strategy</code>
NetShield	Use the NetShield algorithm [24] to select nodes that minimize the spectral radius.	<code>netshield</code>
Degree-based	Select the k healthy nodes with the highest degrees. A degenerate case of PageRank, as discussed in class.	<code>degree_strategy</code>
PageRank-based	Select the k healthy nodes with the highest PageRank scores.	<code>pagerank_strategy</code>

5.2 Data Modeling

To apply the vaccination strategies to real-world network data, we first needed to model the datasets appropriately to the vaccination strategies. We utilized code from previous assignments, specifically code related to network representation, centrality measures, making necessary modifications to fit the current problem.

All networks we deal with our undirected graphs, structured such that each line in them corresponds to a single edge between two nodes id-s. Thus, given each such edge, we extrapolate two directed edges, one in each direction, to represent the undirected nature of the graph.

Dealing With Sinks As PageRank’s convergence mainly relies on the preservation of L1-norm of 1 for the ranks vector, we must deal with nodes that have no outgoing edges, as they would leak rank. We solve this by assuming each sink node has an edge to itself when computing the PageRank scores (i.e. we do not change the structure of the graph, but rather convolute the PageRank computation to account for this).

5.3 Experimental Setup

Social Networks We conduct experiments on four real-world social and peer-to-peer networks, detailed in [Appendix B](#).

Epidemic Propagation Models Configurations We simulate both the IC and SIR models on each network to model different disease dynamics, as discussed in [Section 3.1](#). The configuration parameter ranges are varied based on the network size. We vary the vaccination budget size to evaluate the strategies’ performance under different resource constraints.

Network	Initial Infection Size ($ S $)	Vaccination Budget (k)	Simulations/Configuration
Facebook, LastFM, Gnutella	100	10,30,50,70,90,110,130,150	200
Brightkite	400	10,50,100,150,200,250,300,350	50

Evaluation Metric We evaluate each strategy based on the **average number of healthy nodes** remaining after the epidemic. A higher number of healthy nodes indicates a more effective vaccination strategy. This is averaged across the number of simulations per configuration.

run_experiment The `run_experiment` packs the entire experiment process, including the simulation of the epidemic, the vaccination process, and the evaluation of the results. It is used to run the experiments for each network and vaccination strategy.

6 Experimental Results, Discussion, and Conclusions

Detailed figures for each network under both epidemic models are provided in the [Artifacts Appendix](#) (see Figures 1 and 2).

Our experiments consistently demonstrate that the PageRank-based vaccination strategy significantly reduces the spread of epidemics compared to other heuristics across different networks and epidemic models. Moreover, we note a significant runtime improvement compared to the NetShield algorithm, which requires solving an optimization problem to minimize the spectral radius.

Discussion The effectiveness of the PageRank-based strategy can be attributed to its ability to identify nodes that are not only highly connected but also central in the network’s structure. These nodes have a higher potential to influence a large portion of the network [17]. An additional ability of PageRank that may attribute to its success is its imitation of a random walk, which is closely tied to the spread of diseases in networks.

While NetShield is designed to minimize the spectral radius, our results suggest that PageRank centrality offers a more practical and computationally efficient heuristic.

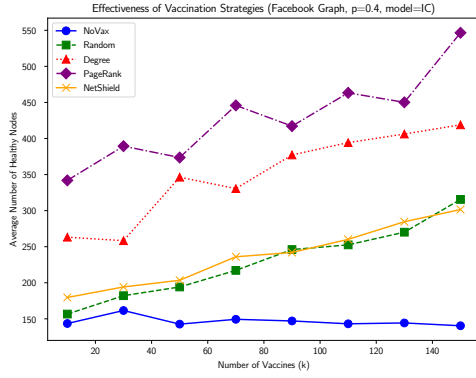
Conclusions We have shown that using PageRank centrality as a heuristic for node vaccination in the Data-Aware Vaccination problem is highly effective in controlling the spread of epidemics in networks. Our extensive experiments on real-world networks and epidemic models validate our hypothesis and demonstrate that the PageRank-based strategy outperforms other heuristics.

Future work includes exploring dynamic vaccination strategies that adapt to the evolving state of the epidemic and investigating the combination of PageRank with other centrality measures to further enhance vaccination effectiveness.

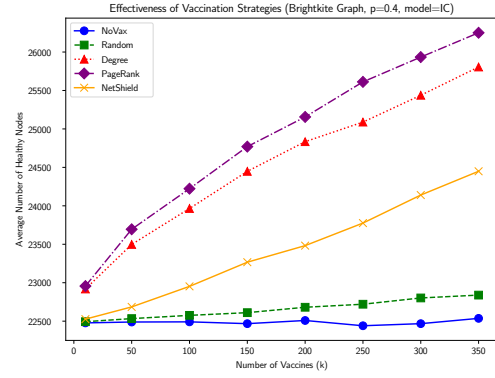
A Artifacts

In this appendix, we present the experimental results in the form of two composite figures, each containing four subfigures corresponding to the different networks analyzed under each epidemic model.

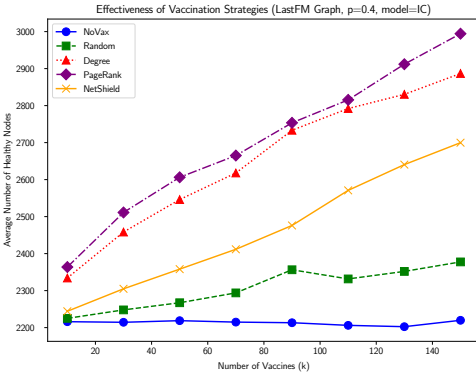
A.1 Independent Cascade Model Results (COVID-19-like)



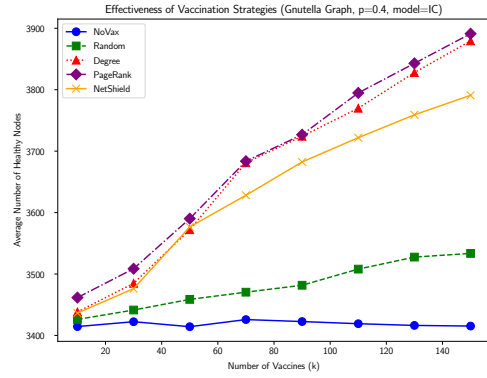
(a) Facebook Network



(b) Brightkite Network



(c) LastFM Network



(d) Gnutella Network

Figure 1: Effectiveness of Vaccination Strategies under the Independent Cascade Model ($p=0.4$)

A.2 SIR Model Results (Ebola-like)

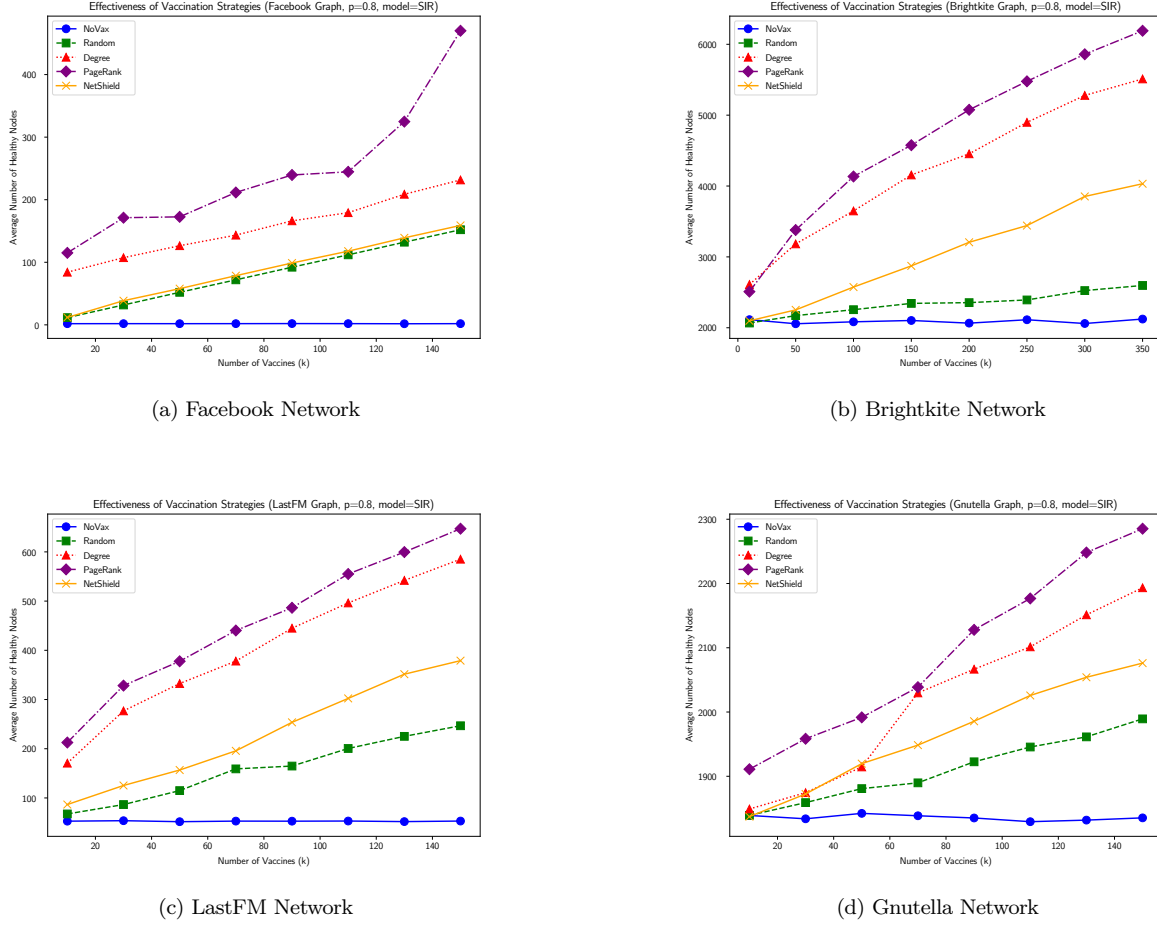


Figure 2: Effectiveness of Vaccination Strategies under the SIR Model ($p=0.8$, $\delta=0.1$)

B Datasets

We used the following real-world network datasets in our experiments, taken from the [14, Stanford Network Analysis Project (SNAP)]:

Dataset	Description	Nodes	Edges
Facebook Social Network [6, 15]	A network representing friendships on Facebook.	4,039	88,234
Brightkite Friendship Network [3, 4]	A network from the Brightkite location-based social networking platform. We used the friendship network for our experiments.	58,228	214,078
LastFM Music Social Network [12, 22]	A network representing friendships on LastFM, a music recommendation platform.	7,624	27,806
Gnutella Peer-to-Peer Network [1, 13, 21]	A network of Gnutella peer-to-peer file sharing connections. It serves as a model for online virus spread, expanding the scope of our experiments.	6,301	20,777

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